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Dear Editor,

A Unifying Theory of Visual Foraging

We would be grateful if you would consider our manuscript for publication in *Journal of Experimental Psychology: General*.

Visual foraging is a behavioural task where participants sequentially collect multiple targets that are distributed through space: it is an ubiquitous behaviour, found in both humans and non-human animals, and can involve searching through both physical space (such as searching for berries on a bush) and abstract space (such as when retrieving a word from memory). Theoretical approaches to understanding visual foraging therefore have the potential to be highly influential across multiple domains of psychology. In the current manuscript, we present a robust and flexible generative model for this type of spatial-sequential data that allows prediction of participants' selection behaviour on a target-by-target basis.

The starting point of our model is to treat foraging as a sampling-without-replacement process in which the probability of selecting any of the potential targets depends on a small number of latent variables. This allows us to specify a fully generative model that can flexibly account for a range of differences between experimental conditions and from one individual to the next. Our approach also maps well onto previously documented foraging behaviours, providing a unifying framework for the field. In addition, it is able to deal with 'outliers' in a principled way: for example, we show that incorporating new parameters in the model to account for absolute direction preferences helps us better explain behaviour of a specific subset of participants who use these types of 'scanning' strategies.

We are submitting this for consideration in Journal of Experimental Psychology: General because we think the manuscript goes beyond an approach to data analysis and into a more important theoretical contribution about how we characterise behavioural effects. We are fully committed to publishing transparent, rigorous research, and have made our code and data available and as accessible as possible in order to encourage further theoretical development. Finally, we think that our approach of iteratively applying, evaluating and improving a model has broader applicability for model development in other areas of psychology, making the manuscript suitable for a general audience.

We suggest that Timothy Vickery and David Kellen may be suitable Associate Editors. Below, we suggest a number of potential reviewers who are all experts in the general field of visual foraging and search. None of these reviewers are affiliated with the authors in any way.

- Beatriz Gil-Gómez de Liaño (bgil.gomezdelianno@uam.es)
- Árni Kristjánsson (ak@hi.is)

- Joe MacInnes (william.macinnes@swansea.ac.uk)
- Carlo De Lillo (cdl2@le.ac.uk)
- Jeremy Wolfe (jwolfe@bwh.harvard.edu)

We look forward to your decision and thank you in advance for your consideration,

Yours sincerely,

Anna Hughes